

The SSIV second stage is a regression of Y_1 on $\hat{W}_{12}$. The correct asymptotic distribution for this estimator is derived in Inoue and Solon (2009), who show that the distribution presented in Angrist and Krueger (1992) requires the assumption that $Z'_1 Z_1 = Z'_2 Z_2$ (as would be true if the marginal distribution of the instruments and covariates is fixed in repeated samples). It's worth noting, however, that the limiting distributions of SSIV and 2SLS are the same when the coefficient on the endogenous variable is zero. The standard errors for this special case are simple to construct and probably provide a reasonably good approximation to the general case.¹⁸

4.4 IV with Heterogeneous Potential Outcomes

The discussion of IV up to this point postulates a constant causal effect. In the case of a dummy variable such as veteran status, this means $Y_{1i} - Y_{0i} = \rho$ for all i , while with a multivalued treatment such as schooling, this means $Y_{si} - Y_{s-1,i} = \rho$ for all s and all i . Both are highly stylized views of the world, especially the multivalued case, which imposes linearity as well as homogeneity. To focus on one thing at a time in a heterogeneous effects model, we start with a zero-one causal variable, like a treatment dummy. In this context, we'd like to allow for treatment effect heterogeneity, in other words, a distribution of causal effects across individuals.

¹⁸This shortcut formula uses the standard errors from the manual SSIV second stage. The correct asymptotic covariance matrix formula, from Inoue and Solon (2005), is

$$\{B'[(\sigma_{11} + \kappa \Gamma' \Sigma_{22} \Gamma)A]^{-1}B\}^{-1},$$

where $B = \text{plim}\left(\frac{Z'_2 W_2}{N_2}\right) = \text{plim}\left(\frac{Z'_1 W_1}{N_1}\right)$, $A = \text{plim}\left(\frac{Z'_1 Z_1}{N_1}\right) = \text{plim}\left(\frac{Z_2 Z_2}{N_2}\right)$, $\text{plim}\left(\frac{N_2}{N_1}\right) = \kappa$, σ_{11} is the variance of the reduced-form residual in data set 1, and Σ_{22} is the variance of the first-stage residual in data set 2. In principle, these pieces are easy enough to calculate. Other approaches to SSIV inference include those of Dee and Evans (2003), who calculate standard errors for just-identified models using the delta method, and Bjorklund and Jantti (1997), who use a bootstrap.

Why is treatment effect heterogeneity important? The answer lies in the distinction between the two types of validity that characterize a research design. *Internal validity* is the question of whether a given design successfully uncovers causal effects for the population being studied. A randomized clinical trial, or for that matter a good IV study, has a strong claim to internal validity. *External validity* is the predictive value of the study's findings in a different context. For example, if the study population in a randomized trial is especially likely to benefit from treatment, the resulting estimates may have little external validity. Likewise, draft lottery estimates of the effects of conscription for service in the Vietnam era need not be a good measure of the consequences of voluntary military service. An econometric framework with heterogeneous treatment effects helps us to assess both the internal and external validity of IV estimates.¹⁹

4.4.1 Local Average Treatment Effects

In an IV framework, the engine that drives causal inference is the instrument z_i , but the variable of interest is still D_i . This feature of the IV setup leads us to adopt a generalized potential outcomes concept, indexed against both instruments and treatment status. Let $Y_i(d, z)$ denote the potential outcome of individual i were this person to have treatment status $D_i = d$ and instrument value $z_i = z$. This tells us, for example, what the earnings of i would be given alternative combinations of veteran status and draft eligibility status. The causal effect of veteran status given i 's realized draft eligibility status is $Y_i(1, z_i) - Y_i(0, z_i)$, while the causal effect of draft eligibility status given i 's veteran status is $Y_i(D_i, 1) - Y_i(D_i, 0)$.

We can think of instrumental variables as initiating a causal chain where the instrument z_i affects the variable of interest, D_i , which in turn affects outcomes, Y_i . To make this precise, we introduce notation to express the idea that the instrument

¹⁹The distinction between internal and external validity has a long history in social science. See, for example, the chapter-length discussion in Shadish, Cook, and Campbell (2002), the successor to a classic text on research methods by Campbell and Stanley (1963).

has a causal effect on D_i . Let D_{1i} be i 's treatment status when $Z_i = 1$, while D_{0i} is i 's treatment status when $Z_i = 0$. Observed treatment status is therefore

$$D_i = D_{0i} + (D_{1i} - D_{0i})Z_i = \pi_0 + \pi_{1i}Z_i + \xi_i. \quad (4.4.1)$$

In random coefficients notation, $\pi_0 \equiv E[D_{0i}]$ and $\pi_{1i} \equiv (D_{1i} - D_{0i})$, so π_{1i} is the heterogeneous causal effect of the instrument on D_i . As with potential outcomes, only one of the potential treatment assignments, D_{1i} and D_{0i} , is ever observed for any one person. In the draft lottery example, D_{0i} tells us whether i would serve in the military if he drew a high (draft-ineligible) lottery number, while D_{1i} tells us whether i would serve if he drew a low (draft-eligible) lottery number. We get to see one or the other of these potential assignments depending on Z_i . The average causal effect of Z_i on D_i is $E[\pi_{1i}]$.

The first assumption in the heterogeneous effects framework is that the instrument is as good as randomly assigned: it is independent of the vector of potential outcomes and potential treatment assignments. Formally, this can be written as

$$[Y_i(d, z); \forall d, z], D_{1i}, D_{0i}] \perp\!\!\!\perp Z_i. \quad (4.4.2)$$

This *independence assumption* is sufficient for a causal interpretation of the reduced form, that is, the regression of Y_i on Z_i . Specifically,

$$\begin{aligned} E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] \\ &= E[Y_i(D_{1i}, 1)|Z_i = 1] - E[Y_i(D_{0i}, 0)|Z_i = 0] \\ &= E[Y_i(D_{1i}, 1) - Y_i(D_{0i}, 0)], \end{aligned}$$

the causal effect of the instrument on Y_i . Independence also means that

$$\begin{aligned} E[D_i|Z_i = 1] - E[D_i|Z_i = 0] &= E[D_{1i}|Z_i = 1] - E[D_{0i}|Z_i = 0] \\ &= E[D_{1i} - D_{0i}]. \end{aligned}$$

In other words, the first stage from our earlier discussion of 2SLS captures the causal effect of Z_i on D_i .

The second key assumption in the heterogeneous effects framework is the presumption that $Y_i(d, z)$ is a function only of d . To be specific, while draft eligibility clearly affects veteran status, an individual's potential earnings as a veteran or

nonveteran are assumed to be unchanged by draft eligibility status. In general, the claim that an instrument operates through a single known causal channel is called an *exclusion restriction*. Formally, the exclusion restriction says that

$$y_i(d, 0) = y_i(d, 1) \text{ for } d = 0, 1.$$

In a linear model with constant effects, the exclusion restriction is expressed by the omission of instruments from the causal equation of interest and by saying that $E[z_i \eta_i] = 0$ in an equation like (4.1.14) in section 4.1. It's worth noting however, that the traditional error term notation used for simultaneous equations models doesn't lend itself to a clear distinction between independence and exclusion. We need z_i and η_i to be uncorrelated, but the reasoning that lies behind this assumption is unclear until we consider the independence and exclusion restrictions as distinct propositions.

The exclusion restriction fails for draft lottery instruments if men with low draft lottery numbers were affected in some way other than through an increased likelihood of military service. For example, Angrist and Krueger (1992) looked for an association between draft lottery numbers and schooling. Their idea was that educational draft deferments could have led men with low lottery numbers to stay in college longer than they would have otherwise desired. If so, draft lottery numbers are correlated with earnings for at least two reasons: an increased likelihood of military service and an increased likelihood of college attendance. The fact that the lottery number is randomly assigned (and therefore satisfies the independence assumption) does not make this possibility less likely. The exclusion restriction is distinct from the claim that the instrument is (as good as) randomly assigned. Rather, it is a claim about a unique channel for causal effects of the instrument.²⁰

²⁰As it turns out, there is not much of a relationship between schooling and lottery numbers in the Angrist and Krueger (1992) data, probably because educational deferments were phased out during the lottery period. On the other hand, in a recent paper, Angrist and Chen (2007) argue that Vietnam veterans end up with more schooling because of veterans benefits (known as the GI Bill). Extra schooling via the GI Bill does not violate the exclusion restriction because veterans' benefits are a downstream consequence of military service.

Using the exclusion restriction, we can define potential outcomes indexed solely against treatment status using the single index (Y_{1i}, Y_{0i}) notation we have been using all along. In particular,

$$\begin{aligned} Y_{1i} &\equiv Y_i(1, 1) = Y_i(1, 0); \\ Y_{0i} &\equiv Y_i(0, 1) = Y_i(0, 0). \end{aligned} \quad (4.4.3)$$

The observed outcome, Y_i , can therefore be written in terms of potential outcomes as:

$$\begin{aligned} Y_i &= Y_i(0, Z_i) + [Y_i(1, Z_i) - Y_i(0, Z_i)]D_i \\ &= Y_{0i} + (Y_{1i} - Y_{0i})D_i. \end{aligned} \quad (4.4.4)$$

Random coefficients notation for this is

$$Y_i = \alpha_0 + \rho_i D_i + \eta_i,$$

a compact version of (4.4.4) with $\alpha_0 \equiv E[Y_{0i}]$ and $\rho_i \equiv Y_{1i} - Y_{0i}$.

A final assumption needed for heterogeneous IV models is that either $\pi_{1i} \geq 0$ for all i or $\pi_{1i} \leq 0$ for all i . This *monotonicity assumption*, introduced by Imbens and Angrist (1994), means that while the instrument may have no effect on some people, all those who are affected are affected in the same way. In other words, either $D_{1i} \geq D_{0i}$ or $D_{1i} \leq D_{0i}$ for all i . In what follows, we assume monotonicity holds with $D_{1i} \geq D_{0i}$. In the draft lottery example, this means that although draft eligibility may have had no effect on the probability of military service for some men, there is no one who was actually kept out of the military by being draft eligible. Without monotonicity, IV estimators are not guaranteed to estimate a weighted average of the underlying individual causal effects, $Y_{1i} - Y_{0i}$.

Given the exclusion restriction, the independence of instruments and potential outcomes, the existence of a first stage, and monotonicity, the Wald estimand can be interpreted as the effect of veteran status on those whose treatment status can be changed by the instrument. This parameter is called the local average treatment effect (LATE; Imbens and Angrist, 1994). Here is a formal statement:

Theorem 4.4.1 *The LATE Theorem. Suppose*

(A1, Independence) $\{Y_i(D_{1i}, 1), Y_i(D_{0i}, 0), D_{1i}, D_{0i}\} \perp\!\!\!\perp Z_i$;

(A2, Exclusion) $Y_i(d, 0) = Y_i(d, 1) \equiv Y_{di}$ for $d = 0, 1$;

(A3, First stage) $E[D_{1i} - D_{0i}] \neq 0$;

(A4, Monotonicity) $D_{1i} - D_{0i} \geq 0 \forall i$, or vice versa;

Then

$$\begin{aligned} \frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[D_i|Z_i = 1] - E[D_i|Z_i = 0]} &= E[Y_{1i} - Y_{0i}|D_{1i} > D_{0i}] \\ &= E[\rho_i|\pi_{1i} > 0]. \end{aligned}$$

Proof. Use the exclusion restriction to write $E[Y_i|Z_i = 1] = E[Y_{0i} + (Y_{1i} - Y_{0i})D_i|Z_i = 1]$, which equals $E[Y_{0i} + (Y_{1i} - Y_{0i})D_{1i}]$ by independence.²¹ Likewise $E[Y_i|Z_i = 0] = E[Y_{0i} + (Y_{1i} - Y_{0i})D_{0i}]$, so the numerator of the Wald estimator is $E[(Y_{1i} - Y_{0i})(D_{1i} - D_{0i})]$. Monotonicity means $D_{1i} - D_{0i}$ equals one or zero, so

$$E[(Y_{1i} - Y_{0i})(D_{1i} - D_{0i})] = E[Y_{1i} - Y_{0i}|D_{1i} > D_{0i}]P[D_{1i} > D_{0i}].$$

A similar argument shows

$$E[D_i|Z_i = 1] - E[D_i|Z_i = 0] = E[D_{1i} - D_{0i}] = P[D_{1i} > D_{0i}].$$

This theorem says that an instrument that is as good as randomly assigned, affects the outcome through a single known channel, has a first stage, and affects the causal channel of interest only in one direction can be used to estimate the average causal effect on the affected group. Thus, IV estimates of effects of military service using the draft lottery capture the effect of military service on men who served because they were draft eligible but who would not otherwise have served. This excludes volunteers and men who were exempted from military service for medical reasons, but it includes men for whom draft policy was binding.

How useful is LATE? No theorem answers this question, but it's always worth discussing. Part of the interest in the effects of

²¹Note that the statement of independence in A1 has been simplified from (4.4.2) to cover only those values of $Y_i(d, Z)$ that we might see, specifically, $Y_i(D_{Z_i}, Z_i)$.

Vietnam-era service has to do with the question of whether veterans (especially conscripts) were adequately compensated for their service. Internally valid draft lottery estimates answer this question. Draft lottery estimates of the effects of Vietnam-era conscription may also be relevant for discussions of any future conscription policy. On the other hand, while draft lottery instruments produce internally valid estimates of the causal effect of Vietnam-era conscription, the external validity—that is, the predictive value of these estimates for military service in other times and places—is not directly addressed by the IV framework. There is nothing in IV formulas to explain *why* Vietnam-era service affects earnings; for that, you need a theory.²²

You might wonder why we need monotonicity for the LATE theorem, an assumption that plays no role in the traditional simultaneous equations framework with constant effects. A failure of monotonicity means the instrument pushes some people into treatment while pushing others out. Angrist, Imbens, and Rubin (1996) call the latter group *defiers*. Defiers complicate the link between LATE and the reduced form. To see why, go back to the step in the proof of the LATE theorem that shows the reduced form is

$$E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] = E[(Y_{1i} - Y_{0i})(D_{1i} - D_{0i})].$$

Without monotonicity, this is equal to

$$\begin{aligned} &E[Y_{1i} - Y_{0i}|D_{1i} > D_{0i}]P[D_{1i} > D_{0i}] \\ &\quad - E[Y_{1i} - Y_{0i}|D_{1i} < D_{0i}]P[D_{1i} < D_{0i}]. \end{aligned}$$

We might therefore have a scenario where treatment effects are positive for everyone yet the reduced form is zero because effects on compliers are canceled out by effects on defiers. This doesn't come up in a constant effects model because the reduced form is always the constant effect times the first

²²Angrist (1990) interprets draft lottery estimates as the penalty for lost labor market experience. This suggests draft lottery estimates should have external validity for the effects of conscription in other periods, a conjecture born out by the results for World War II draftees in Angrist and Krueger (1994).

stage, regardless of whether the first stage includes defiant behavior.²³

A deeper understanding of LATE can be had by linking it to a workhorse of contemporary econometrics, the latent index model for dummy endogenous variables such as assignment to treatment. Latent index models describe individual choices as being determined by a comparison of partly observed and partly unknown (“latent”) utilities and costs (see, e.g., Heckman, 1978). Typically, these unobservables are thought of as being related to outcomes, in which case the treatment variable is said to be endogenous (though it is not really endogenous in a simultaneous equations sense). For example, we can model veteran status as

$$D_i = \begin{cases} 1 & \text{if } \gamma_0 + \gamma_1 Z_i > v_i \\ 0 & \text{otherwise} \end{cases}$$

where v_i is a random factor involving unobserved costs and benefits of military service assumed to be independent of Z_i . This latent index model characterizes potential treatment assignments as

$$D_{0i} = 1[\gamma_0 > v_i] \text{ and } D_{1i} = 1[\gamma_0 + \gamma_1 > v_i].$$

Note that in this model, monotonicity is automatically satisfied since γ_1 is a constant. Assuming $\gamma_1 > 0$, LATE can be written

$$E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}] = E[Y_{1i} - Y_{0i} | \gamma_0 + \gamma_1 > v_i > \gamma_0],$$

which is a function of the latent first-stage parameters, γ_0 and γ_1 , as well as the joint distribution of $Y_{1i} - Y_{0i}$ and v_i . This is not, in general, the same as the unconditional average treatment effect, $E[Y_{1i} - Y_{0i}]$, or the effect on the treated,

²³With a constant effect, ρ ,

$$\begin{aligned} E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}] P[D_{1i} > D_{0i}] - E[Y_{1i} - Y_{0i} | D_{1i} < D_{0i}] P[D_{1i} < D_{0i}] \\ &= \rho \{P[D_{1i} > D_{0i}] - P[D_{1i} < D_{0i}]\} \\ &= \rho \{E[D_{1i} - D_{0i}]\}. \end{aligned}$$

So a zero reduced-form effect means either the first stage is zero or $\rho = 0$.

$E[Y_{1i} - Y_{0i} | D_i = 1]$. We explore the distinction between different average causal effects in the next section.

4.4.2 *The Compliant Subpopulation*

The LATE framework partitions any population with an instrument into a set of three instrument-dependent subgroups, defined by the manner in which members of the population react to the instrument:

Definition 4.4.1 *Compliers.* The subpopulation with $D_{1i} = 1$ and $D_{0i} = 0$.

Always-takers. The subpopulation with $D_{1i} = D_{0i} = 1$.

Never-takers. The subpopulation with $D_{1i} = D_{0i} = 0$.

LATE is the effect of treatment on the population of compliers. The term “compliers” comes from an analogy with randomized trials where some experimental subjects comply with the randomly assigned treatment protocol (e.g., take their medicine) but some do not, while some control subjects obtain access to the experimental treatment even though they are not supposed to. Those who don’t take their medicine when randomly assigned to do so are never-takers, while those who take the medicine even when put into the control group are always-takers. Without adding further assumptions (e.g., constant causal effects), LATE is not informative about effects on never-takers and always-takers because, by definition, treatment status for these two groups is unchanged by the instrument. The analogy between IV and a randomized trial with partial compliance is more than allegorical: IV solves the problem of causal inference in a randomized trial with partial compliance. This important point merits a separate subsection, below.

Before turning to this important special case, we make a few general points. First, the average causal effect on compliers is not usually the same as the average treatment effect on the treated. From the simple fact that $D_i = D_{0i} + (D_{1i} - D_{0i})Z_i$, we learn that the treated population consists of two nonoverlapping groups. By monotonicity, we cannot have both $D_{0i} = 1$

and $D_{1i} - D_{0i} = 1$, since $D_{0i} = 1$ implies $D_{1i} = 1$. The treated therefore have *either* $D_{0i} = 1$ or $D_{1i} - D_{0i} = 1$ and $Z_i = 1$, and hence D_i can be written as the sum of two mutually exclusive dummies, D_{i0} and $(D_{1i} - D_{0i})Z_i$. In other words, the treated consist of either always-takers or of compliers with the instrument switched on. Since the instrument is as good as randomly assigned, compliers with the instrument switched on are representative of all compliers. From here we get

$$\begin{aligned}
 & \underbrace{E[Y_{1i} - Y_{0i} | D_i = 1]}_{\text{Effect on the treated}} \\
 &= \underbrace{E[Y_{1i} - Y_{0i} | D_{0i} = 1]}_{\text{Effect on always-takers}} P[D_{0i} = 1 | D_i = 1] \\
 &+ \underbrace{E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}]}_{\text{Effect on compliers}} P[D_{1i} > D_{0i}, Z_i = 1 | D_i = 1].
 \end{aligned}
 \tag{4.4.5}$$

Since $P[D_{0i} = 1 | D_i = 1]$ and $P[D_{1i} > D_{0i}, Z_i = 1 | D_i = 1]$ add up to one, this means that the effect of treatment on the treated is a weighted average of effects on always-takers and compliers.

Likewise, LATE is not the average causal effect of treatment on the nontreated, $E[Y_{1i} - Y_{0i} | D_i = 0]$. In the draft lottery example, the average effect on the nontreated is the average causal effect of military service on the population of nonveterans from Vietnam-era cohorts. The average effect of treatment on the nontreated is a weighted average of effects on never-takers and compliers. In particular,

$$\begin{aligned}
 & \underbrace{E[Y_{1i} - Y_{0i} | D_i = 0]}_{\text{Effect on the nontreated}} \\
 &= \underbrace{E[Y_{1i} - Y_{0i} | D_{1i} = 0]}_{\text{Effect on never-takers}} P[D_{1i} = 0 | D_i = 0] \\
 &+ \underbrace{E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}]}_{\text{Effect on compliers}} P[D_{1i} > D_{0i}, Z_i = 0 | D_i = 0],
 \end{aligned}
 \tag{4.4.6}$$

where we use the fact that, by monotonicity, those with $D_{1i} = 0$ must be never-takers.

Finally, averaging (4.4.5) and (4.4.6) using

$$E[Y_{1i} - Y_{0i}] = E[Y_{1i} - Y_{0i} | D_i = 1]P[D_i = 1] \\ + E[Y_{1i} - Y_{0i} | D_i = 0]P[D_i = 0]$$

shows the unconditional average treatment effect to be a weighted average of effects on compliers, always-takers, and never-takers. Of course, this is a conclusion we could have reached directly given monotonicity and definition (4.4.1).

Because an IV is not directly informative about effects on always-takers and never-takers, instruments do not usually capture the average causal effect on all of the treated or on all of the nontreated. There are important exceptions to this rule, however: instrumental variables that allow no always-takers or no never-takers. Although this scenario is not typical, it is an important special case. One example is the twins instrument for fertility, used by Rosenzweig and Wolpin (1980), Bronars and Grogger (1994), Angrist and Evans (1998), and Angrist, Lavy, and Schlosser (2006). Another is Oreopoulos's (2006) recent study using changes in compulsory attendance laws as instruments for schooling in Britain.

To see how this special case works with twins instruments, let τ_i be a dummy variable indicating multiple second births. Angrist and Evans (1998) used this instrument to estimate the causal effect of having three children on earnings in the population of women with at least two children. The third child is especially interesting because reduced fertility for American wives in the 1960s and 1970s meant a switch from three children to two. Multiple second births provide quasi-experimental variation on this margin. Let y_{0i} denote potential earnings if a woman has only two children while y_{1i} denotes her potential earnings if she has three, an event indicated by D_i . Assuming that τ_i is as good as randomly assigned, that fertility increases by at most one child in response to a multiple birth, and that multiple births affect outcomes only by increasing fertility, LATE using the twins instrument τ_i is also $E[Y_{1i} - Y_{0i} | D_i = 0]$, the average causal effect on women who are not treated (i.e., have two children only). This is because all women who have a multiple second birth end up with three

children, that is, there are no never-takers in response to the twins instrument.

Oreopoulos (2006) also uses IV to estimate an average causal effect of treatment on the nontreated. His study estimates the economic returns to schooling using an increase in the British compulsory attendance age from 14 to 15. Compliance with the Britain's new compulsory attendance law was near perfect, though many teens would previously have dropped out of school at age 14. The causal effect of interest in this case is the earnings premium for an additional year of high school. Finishing this year can be thought of as the treatment. Since everybody in Oreopoulos's British sample finished an additional year when compulsory schooling laws were made stricter, there are no never-takers. Oreopoulos's IV strategy therefore captures the average causal effect of obtaining one more year of high school on all those who leave school at 14. This turns on the fact that British teens are remarkably law-abiding people—Oreopoulos's IV strategy wouldn't estimate the effect of treatment on the nontreated in, say, Israel, where teenagers get more leeway when it comes to compulsory school attendance. Israeli econometricians using changes in compulsory attendance laws as instruments must therefore make do with LATE.

4.4.3 *IV in Randomized Trials*

The language of the LATE framework is based on an analogy between IV and randomized trials. But some instruments really do come from randomized trials. If the instrument is a randomly assigned offer of treatment, then LATE is the effect of treatment on those who comply with the offer but are not treated otherwise. An especially important case is when the instrument is generated by a randomized trial with one-sided noncompliance. In many randomized trials, participation is voluntary among those randomly assigned to receive treatment. On the other hand, no one in the control group has access to the experimental intervention. Since the group that receives (i.e., complies with) the assigned treatment is a self-selected subset of those offered treatment, a

comparison between those actually treated and the control group is misleading. The selection bias in this case is almost always positive: those who take their medicine in a randomized trial tend to be healthier; those who take advantage of randomly assigned economic interventions such as training programs tend to earn more anyway.

IV using randomly assigned treatment intended as an instrumental variable for treatment received solves this sort of compliance problem. Moreover, LATE is the effect of treatment on the treated in this case. Suppose the instrument z_i is a dummy variable indicating random assignment to a treatment group, while D_i is a dummy indicating whether treatment was actually received. In practice, because of noncompliance, D_i is not equal to z_i . An example is the randomized evaluation of the JTPA training program, where only 60 percent of those assigned to be trained received training, while roughly 2 percent of those assigned to the control group received training anyway (Bloom et al., 1997; see also section 7.2.1). Noncompliance in the JTPA arose from lack of interest among participants and the failure of program operators to encourage participation. Since the compliance problem in this case was largely confined to the treatment group, LATE using random assignment, z_i , as an instrument for treatment received, D_i , is the effect of treatment on the treated.

The use of IV to solve compliance problems is illustrated in table 4.4.1, which presents results from the JTPA experiment. The outcome variable of primary interest in the JTPA experiment is total earnings in the 30-month period after random assignment. Columns 1 and 2 of the table show the difference in earnings between those who were trained and those who were not (the OLS estimates in column 2 are from a regression model that adjusts for a number of individual characteristics measured at the beginning of the experiment). The contrast reported in columns 1 and 2 is on the order of \$4,000 for men and \$2,200 for women, in both cases a large treatment difference that amounts to about 20 percent of average earnings. But these estimates are misleading because they compare individuals according to D_i , the actual treatment received. Since individuals assigned to the treatment group were free to

TABLE 4.4.1

Results from the JTPA experiment: OLS and IV estimates of training impacts

	Comparisons by Training Status (OLS)		Comparisons by Assignment Status (ITT)		Instrumental Variable Estimates (IV)	
	Without Covariates (1)	With Covariates (2)	Without Covariates (3)	With Covariates (4)	Without Covariates (5)	With Covariates (6)
A. Men	3,970 (555)	3,754 (536)	1,117 (569)	970 (546)	1,825 (928)	1,593 (895)
B. Women	2,133 (345)	2,215 (334)	1,243 (359)	1,139 (341)	1,942 (560)	1,780 (532)

Notes: Authors' tabulation of JTPA study data. The table reports OLS, ITT, and IV estimates of the effect of subsidized training on earnings in the JTPA experiment. Columns 1 and 2 show differences in earnings by training status; columns 3 and 4 show differences by random-assignment status. Columns 5 and 6 report the result of using random-assignment status as an instrument for training. The covariates used for columns 2, 4, and 6 are high school or GED, black, Hispanic, married, worked less than 13 weeks in past year, AFDC (for women), plus indicators for the JTPA service strategy recommended, age group, and second follow-up survey. Robust standard errors are shown in parentheses. There are 5,102 men and 6,102 women in the sample.

decline (and 40 percent did so), this comparison throws away the random assignment unless the decision to comply was itself independent of potential outcomes. This seems unlikely.

Columns 3 and 4 of table 4.4.1 compare individuals according to whether they were *offered* treatment. In other words, this comparison is based on randomly assigned z_i . In the language of clinical trials, the contrast in columns 3 and 4 is known as an *intention-to-treat* (ITT) effect. The ITT effects in the table are on the order of \$1,200 (somewhat less with covariates). Since z_i was randomly assigned, ITT effects have a causal interpretation: they tell us the causal effect of the offer of treatment, building in the fact that many of those offered have declined to participate. For this reason, the ITT effect is too small relative to the average causal effect on those who were in fact treated. Columns 5 and 6 put the pieces together and give us the most interesting effect: ITT divided by the difference in compliance rates between treatment and control groups as originally assigned (about .6). These figures, roughly \$1,800, measure the effect of treatment on the treated.

How do we know that ITT divided by compliance is the effect of treatment on the treated? We can recognize ITT as the reduced-form effect of the randomly assigned offer of treatment, our instrument in this case. The compliance rate is the first stage associated with this instrument, and the Wald estimand, as always, is the reduced form divided by the first stage. In general, this equals LATE, but because we have (almost) no always-takers, the treated population consists (almost) entirely of compliers. The IV estimates in columns 5 and 6 of table 4.4.1 are therefore consistent estimates of the effect of treatment on the treated.

This conclusion is important enough that it warrants an alternative derivation. To the best of our knowledge, the first person to point out that the IV formula can be used to estimate the effect of treatment on the treated in a randomized trial with one-sided noncompliance was Howard Bloom (1984). Here is Bloom's result with a simple direct proof.

Theorem 4.4.2 *The Bloom Result. Suppose the assumptions of the LATE theorem hold, and $E[D_i|Z_i = 0] = P[D_i = 1|Z_i = 0] = 0$. Then*

$$\frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{P[D_i = 1|Z_i = 1]} = E[Y_{1i} - Y_{0i}|D_i = 1].$$

Proof. $E[Y_i|Z_i = 1] = E[Y_{0i}|Z_i = 1] + E[(Y_{1i} - Y_{0i})D_i|Z_i = 1]$, while $E[Y_i|Z_i = 0] = E[Y_{0i}|Z_i = 0]$ because $Z_i = 0$ implies $D_i = 0$. Therefore

$$E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] = E[(Y_{1i} - Y_{0i})D_i|Z_i = 1],$$

since $E[Y_{0i}|Z_i = 0] = E[Y_{0i}|Z_i = 1]$ by independence. But

$$\begin{aligned} E[(Y_{1i} - Y_{0i})D_i|Z_i = 1] \\ = E[Y_{1i} - Y_{0i}|D_i = 1, Z_i = 1]P[D_i = 1|Z_i = 1], \end{aligned}$$

while $D_i = 1$ implies $Z_i = 1$, since no one with $Z_i = 0$ is treated. Hence, $E[Y_{1i} - Y_{0i}|D_i = 1, Z_i = 1] = E[Y_{1i} - Y_{0i}|D_i = 1]$.

In addition to telling us how to analyze randomized trials with noncompliance, the LATE framework opens the

door to cleverly designed randomized experiments in settings where it's impossible or unethical to compel treatment compliance. A creative example from the field of criminology is the Minneapolis Domestic Violence Experiment (MDVE). The MDVE was a pioneering effort to determine the best police response to domestic violence (Sherman and Berk, 1984). In general, police use a number of strategies in response to a domestic violence call. These include referral to counseling, separation orders, and arrest. A vigorous debate swirls around the question of whether a hard-line response—arrest and at least temporary incarceration—is productive, especially in view of the fact that domestic assault charges are frequently dropped.

As a result of this debate, the city of Minneapolis authorized a randomized trial where the police response to a domestic disturbance call was determined in part by random assignment. The research design used randomly shuffled color-coded report forms telling the responding officers to arrest some perpetrators while referring others to counseling or merely separating the parties. In practice, however, the police were free to overrule the random assignment. For example, an especially dangerous or drunk offender was arrested no matter what. As a result, the actual response often deviated from the randomly assigned response, though the two are highly correlated.

Most published analyses of the MDVE data recognize this compliance problem and focus on ITT effects, that is, they use the original random assignment and not the treatment actually delivered. But the MDVE data can also be used to get the average causal effect on compliers, in this case those who were not arrested because they were randomly assigned to be treated differently but would have been arrested otherwise. The MDVE is analyzed in this spirit in Angrist (2006). Because everyone in the MDVE who was assigned to be arrested was in fact arrested, there are no never-takers. This is an interesting twist and the flip side of the Bloom scenario: here we have $D_{1i} = 1$ for everybody. Consequently, LATE is the effect of treatment on the nontreated, that is,

$$E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}] = E[Y_{1i} - Y_{0i} | D_i = 0],$$

where D_i indicates arrest. The IV estimates using MDVE data show that the average causal effect of arrest is to reduce repeat offenses sharply, in this case, among the subpopulation not arrested.²⁴

4.4.4 *Counting and Characterizing Compliers*

We've seen that, except in special cases, each instrumental variable identifies a unique causal parameter, one specific to the subpopulation of compliers for that instrument. Different valid instruments for the same causal relation therefore estimate different things, at least in principle (an important exception being instruments that allow for perfect compliance on one side or the other). Although different IV estimates are implicitly weighted up by 2SLS to produce a single average causal effect, overidentification testing of the sort discussed in section 4.2.2, where multiple instruments are validated according to whether or not they estimate the same thing, is out the window in a fully heterogeneous world.

Differences in compliant subpopulations might explain variability in treatment effects from one instrument to another. We would therefore like to learn as much as we can about the compliers for different instruments. Moreover, if the compliant subpopulation is similar to other populations of interest, the case for extrapolating estimated causal effects to these other populations is stronger. In this spirit, Acemoglu and Angrist (2000) argue that quarter-of-birth instruments and state compulsory attendance laws (specifically, the minimum schooling required before leaving school in your state of birth) affect essentially the same group of people and for the same reasons. We therefore expect IV estimates of the returns to

²⁴The Krueger (1999) study discussed in chapter 2 also uses IV to analyze data from a randomized trial. Specifically, this study uses randomly assigned class size as an instrument for actual class size with data from the Tennessee STAR experiment. For students in first grade and higher, actual class size differs from randomly assigned class size because parents and teachers moved students around in years after the experiment began. Krueger (1999) also illustrates 2SLS applied to a model with variable treatment intensity, as discussed in section 4.5.3.

schooling from quarter-of-birth and compulsory schooling instruments to be similar. We might also expect the quarter-of-birth estimates to predict the impact of contemporary proposals to strengthen compulsory attendance laws.

On the other hand, if the compliant subpopulations associated with two or more instruments are very different, yet the IV estimates they generate are similar, we might be prepared to adopt homogeneous effects as a working hypothesis. This revives the overidentification idea but puts it at the service of external validity.²⁵ This reasoning is illustrated in a study of the effects of family size on children's education by Angrist, Lavy, and Schlosser (2006). The Angrist, Lavy, and Schlosser study was motivated by the observation that children from larger families typically end up with less education than those from smaller families. A longstanding concern in research on fertility is whether the observed negative correlation between larger families and worse outcomes is causal. As it turns out, IV estimates of the effect of family size using a number of different instruments, each with very different compliant subpopulations, all generate results showing no effect of family size. Angrist, Lavy, and Schlosser (2006) argue that their results point to a common family size effect of zero for just about everybody in the Israeli population they studied.²⁶

We have already seen that the size of a complier group is easy to measure. This is just the Wald first stage, since, given monotonicity, we have

$$\begin{aligned} P[D_{1i} > D_{0i}] &= E[D_{1i} - D_{0i}] \\ &= E[D_{1i}] - E[D_{0i}] \\ &= E[D_i | Z_i = 1] - E[D_i | Z_i = 0]. \end{aligned}$$

We can also tell what proportion of the treated are compliers since, for compliers, treatment status is completely determined

²⁵In fact, maintaining the hypothesis that all instruments in an overidentified model are valid, the traditional overidentification test statistic becomes a formal test for treatment effect homogeneity.

²⁶See also Black, Devereux, and Salvones (2005) for similar results from Norway.

by z_i . Start with the definition of conditional probability:

$$\begin{aligned} P[D_{1i} > D_{0i} | D_i = 1] &= \frac{P[D_i = 1 | D_{1i} > D_{0i}] P[D_{1i} > D_{0i}]}{P[D_i = 1]} \\ &= \frac{P[Z_i = 1](E[D_i | Z_i = 1] - E[D_i | Z_i = 0])}{P[D_i = 1]}. \end{aligned} \quad (4.4.7)$$

The second equality uses the facts that $P[D_i = 1 | D_{1i} > D_{0i}] = P[Z_i = 1 | D_{1i} > D_{0i}]$ and that $P[Z_i = 1 | D_{1i} > D_{0i}] = P[Z_i = 1]$ by independence. In other words, the proportion of the treated who are compliers is given by the first stage, times the probability the instrument is switched on, divided by the proportion treated.

Formula (4.4.7) is illustrated here by calculating the proportion of veterans who are draft lottery compliers. The ingredients are reported in the first two rows of table 4.4.2. For example, for white men born in 1950, the first stage is .159, the probability of draft eligibility is $\frac{195}{365}$, and the marginal probability of treatment is .267. From these statistics, we compute that the compliant subpopulation is .32 of the veteran population in this group. The proportion of veterans who were draft lottery compliers falls to 20 percent for nonwhite men born in 1950. This is not surprising, since the draft lottery first stage is considerably weaker for nonwhites. The last column of the table reports the proportion of *nonveterans* who would have served if they had been draft eligible. This ranges from about 3 percent of nonwhites to 10 percent of whites, reflecting the fact that most nonveterans were deferred, ineligible, or unqualified for military service.

The effect of compulsory military service is the parameter of primary interest in the Angrist (1990) study, so the fact that draft eligibility compliers are a minority of veterans is not really a limitation of this study. Even in the Vietnam era, most soldiers were volunteers, a little appreciated fact about Vietnam-era veterans. The LATE interpretation of IV estimates using the draft lottery highlights the fact that other identification strategies are needed to estimate the effects of military service on volunteers (some of these are implemented in Angrist, 1998).

TABLE 4.4.2
Probabilities of compliance in instrumental variables studies

Source (1)	Endogenous Variable (D) (2)	Instrument (z) (3)	Sample (4)	$P[D = 1]$ (5)	First Stage, $P[D_1 > D_0]$ (6)	$P[z = 1]$ (7)	Compliance Probabilities	
							$P[D_1 > D_0 D = 1]$ (8)	$P[D_1 > D_0 D = 0]$ (9)
Angrist (1990)	Veteran status	Draft eligibility	White men born in 1950	.267	.159	.534	.318	.101
			Non-white men born in 1950	.163	.060	.534	.197	.033
Angrist and Evans (1998)	More than two children	Twins at second birth	Married women aged 21–35 with two or more children in 1980	.381	.603	.008	.013	.966
			First two children are same sex	.381	.060	.506	.080	.048
Angrist and Krueger (1991)	High school graduate	Third- or fourth-quarter birth	Men born between 1930 and 1939	.770	.016	.509	.011	.034
Acemoglu and Angrist (2000)	High school graduate	State requires 11 or more years of school attendance	White men aged 40–49	.617	.037	.300	.018	.068

Notes: The table computes the absolute and relative size of the complier population for a number of instrumental variables. The first stage, reported in column 6, gives the absolute size of the complier group. Columns 8 and 9 show the size of the complier population relative to the treated and untreated populations.

The remaining rows in table 4.4.2 show the size of the compliant subpopulation for the twins and sibling sex composition instruments used by Angrist and Evans (1998) to estimate the effects of childbearing and for the quarter-of-birth instruments and compulsory attendance laws used by Angrist and Krueger (1991) and Acemoglu and Angrist (2000) to estimate the returns to schooling. In each of these studies, the compliant subpopulation is a small fraction of the treated group. For example, less than 2 percent of those who graduated from high school did so because of compulsory attendance laws or by virtue of having been born in a late quarter.

The question of whether a small compliant subpopulation is a cause for worry is context-specific. In some cases, it seems fair to say, “you get what you need.” With many policy interventions, for example, it is a marginal group that is of primary interest, a point emphasized in McClellan, McNeil, and Newhouse’s (1994) landmark IV study of the effects of surgery on heart attack patients. They used the relative distance to cardiac care facilities to construct instruments for whether an elderly heart attack patient was treated with a surgical intervention. Most patients get the same treatment either way, but for some, the proper course of action (or at least the received wisdom as to the proper course of action) is uncertain. In such cases, health care providers or patients opt for a more invasive strategy only if a well-equipped surgical facility is close by. McClellan et al. found little benefit from surgical procedures for this marginal group. Similarly, an increase in the compulsory attendance cut-off to age 18 is clearly irrelevant for the majority of American high school students, but affects some who would otherwise drop out. IV estimates suggest the economic returns to schooling for this marginal group are substantial.

The last column of table 4.4.2 illustrates the special feature of twins instruments alluded to at the end of section 4.4.2. As before, let $D_i = 0$ for women with two children in a sample of women with at least two children and 1 for women who have more than two. Because there are no never-takers in response to the event of a multiple birth—all mothers who have twins at second birth end up with (at least) three children—the

probability of compliance among those with $D_i = 0$ is virtually one (the table shows an entry of .97). LATE is therefore the effect on the nontreated, $E[Y_{1i} - Y_{0i}|D_i = 0]$, in this case.

Unlike the size of the complier group, information on the *characteristics* of compliers seems like a tall order because the compliers cannot be individually identified. Because we can't see both D_{1i} and D_{0i} for each individual, we can't just list those with $D_{1i} > D_{0i}$ and then calculate the distribution of characteristics for this group. Nevertheless, in spite of the fact that compliers cannot be listed or named, it's easy to describe the distribution of complier characteristics. To simplify, we focus here on characteristics, such as race or degree completion, that can be described by dummy variables. In this case, everything we need to know can be learned from variation in the first stage across covariate groups.

Let x_{1i} be a Bernoulli-distributed characteristic, say a dummy indicating college graduates. Are sex composition compliers more or less likely to be college graduates than other women with two children? This question is answered by the following calculation:

$$\begin{aligned} \frac{P[x_{1i} = 1|D_{1i} > D_{0i}]}{P[x_{1i} = 1]} &= \frac{P[D_{1i} > D_{0i}|x_{1i} = 1]}{P[D_{1i} > D_{0i}]} \\ &= \frac{E[D_i|Z_i = 1, x_{1i} = 1] - E[D_i|Z_i = 0, x_{1i} = 1]}{E[D_i|Z_i = 1] - E[D_i|Z_i = 0]}. \quad (4.4.8) \end{aligned}$$

In other words, the relative likelihood a complier is a college graduate is given by the ratio of the first stage for college graduates to the overall first stage.²⁷

²⁷A general method for constructing the mean or other features of the distribution of covariates for compliers uses Abadie's (2003) kappa-weighting scheme. For example,

$$E[X_i|D_{1i} > D_{0i}] = \frac{E[\kappa_i X_i]}{E[\kappa_i]},$$

where

$$\kappa_i = 1 - \frac{D_i(1 - Z_i)}{1 - P(Z_i = 1|X_i)} - \frac{(1 - D_i)Z_i}{P(Z_i = 1|X_i)}.$$

This works because the weighting function, κ_i , "finds compliers," in a sense discussed in section 4.5.2.

TABLE 4.4.3

Complier characteristics ratios for twins and sex composition instruments

Variable	Twins at Second Birth			First Two Children Are Same Sex	
	$P[x_{1i} = 1]$ (1)	$P[x_{1i} = 1 D_{1i} > D_{0i}]$ (2)	$P[x_{1i} = 1 D_{1i} > D_{0i}] / P[x_{1i} = 1]$ (3)	$P[x_{1i} = 1 D_{1i} > D_{0i}]$ (4)	$P[x_{1i} = 1 D_{1i} > D_{0i}] / P[x_{1i} = 1]$ (5)
Age 30 or older at first birth	.0029	.004	1.39	.0023	.995
Black or hispanic	.125	.103	.822	.102	.814
High school graduate	.822	.861	1.048	.815	.998
College graduate	.132	.151	1.14	.0904	.704

Notes: The table reports an analysis of complier characteristics for twins and sex composition instruments. The ratios in columns 3 and 5 give the relative likelihood that compliers have the characteristic indicated at left. Data are from the 1980 census 5 percent sample, including married mothers aged 21–35 with at least two children, as in Angrist and Evans (1998). The sample size is 254,654 for all columns.

This calculation is illustrated in table 4.4.3, which reports compliers' characteristics ratios for age at first birth, non-white race, and degree completion using twins and same-sex instruments. The table was constructed from the Angrist and Evans (1998) extract from the 1980 census containing married women aged 21–35 with at least two children. Twins compliers are much more likely to be over 30 than the average mother in the sample, reflecting the fact that younger women who had a multiple birth were more likely to go on to have additional children anyway (though over-30 first births are rare for all women in the Angrist-Evans sample). Twins compliers are also more educated than the average mother, while sex composition compliers are less educated. This helps to explain the smaller 2SLS estimates generated by twins instruments (reported here in table 4.1.4), since Angrist and Evans (1998) show that the labor supply consequences of childbearing decline with mother's schooling.